

Technologies stack

Starter

 **Laravel**

 **Laravel** Jetstream

Authentication

 **Laravel** Sanctum

 **Laravel** Fortify

UI/UX Interface

 **livewire**

 Blade

 **Metronic**

Debugging

 **Laravel** Telescope

Queues / Jobs

 **Laravel** Horizon

 redis

RESTful API

 Swagger

Database

 PostgreSQL

InDesign

 **Id** InDesign Server

Containerization

 **Laravel** Sail

 docker

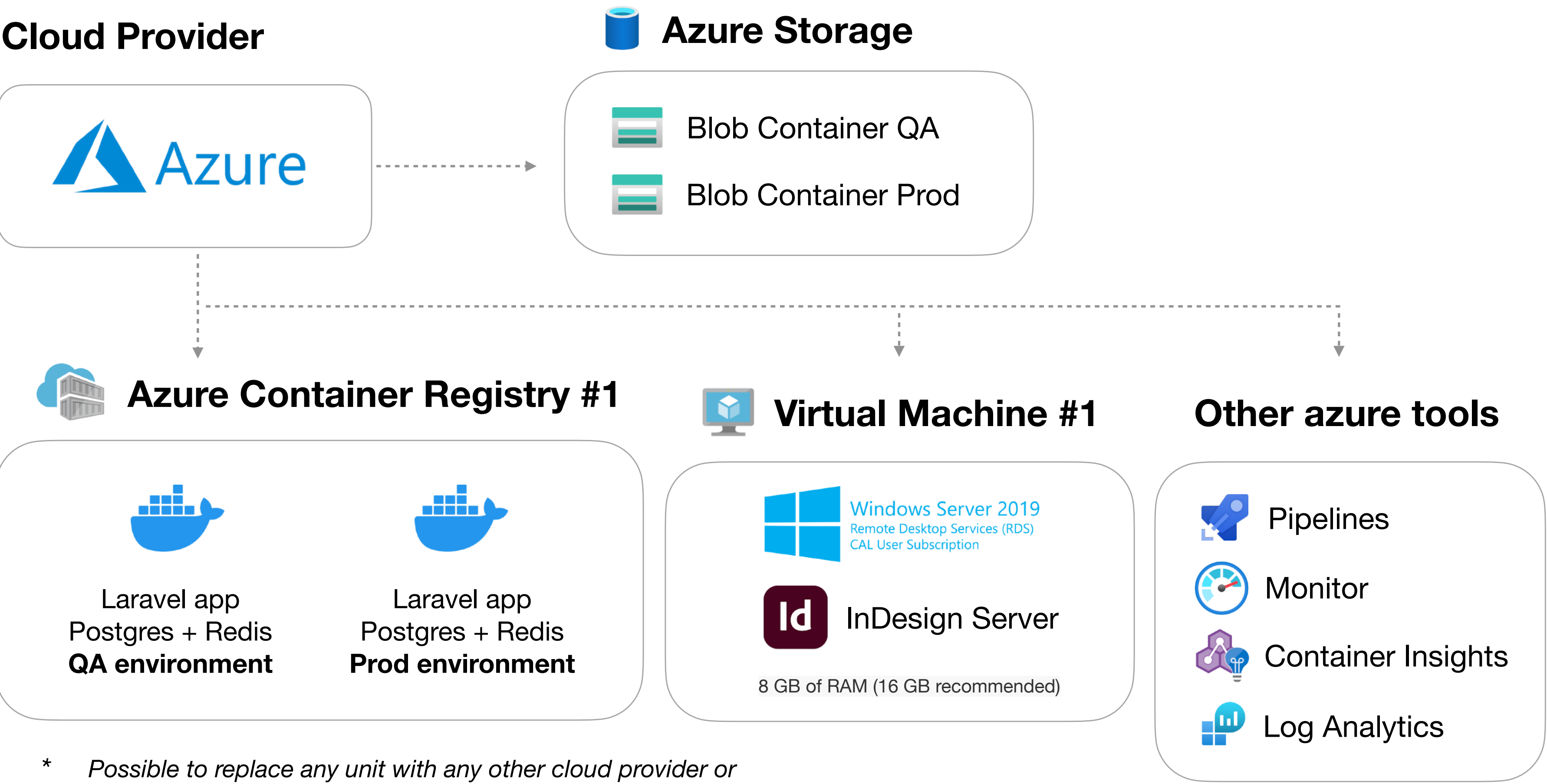
Social Media API

 ayrshare



Server Architecture

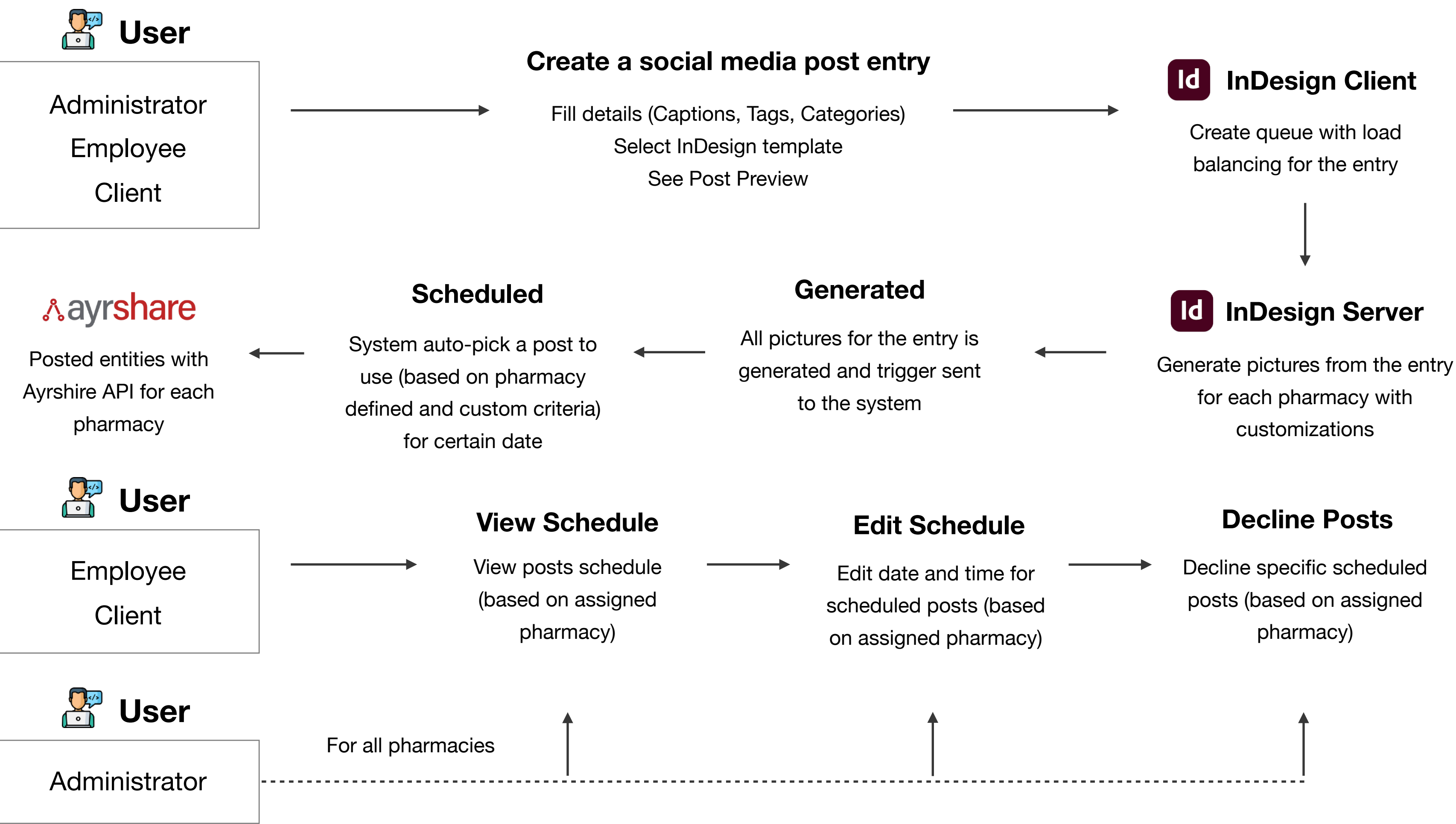
Cloud Provider



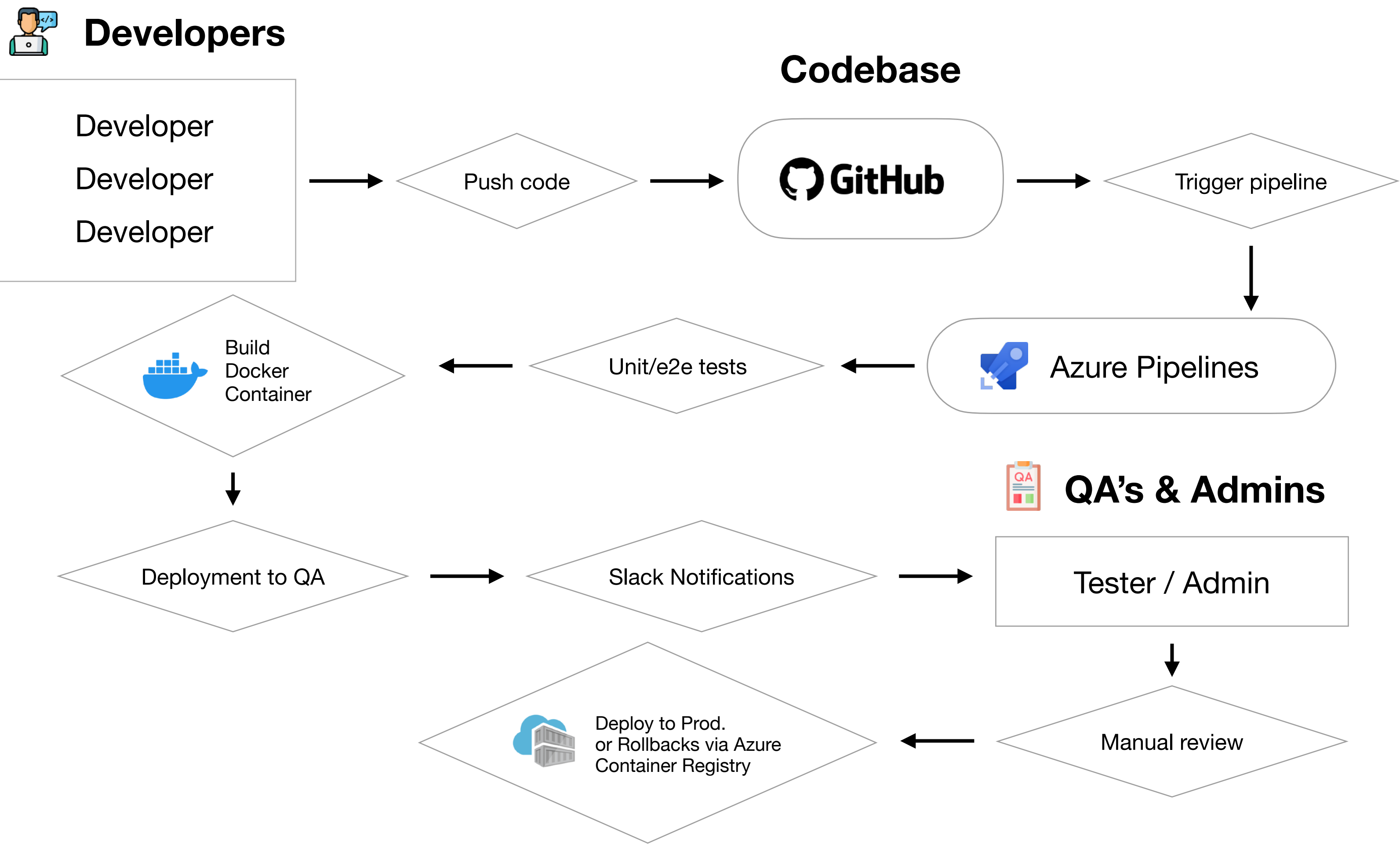
* Possible to replace any unit with any other cloud provider or in-Premise machines to lower monthly expenses



Workflow scenarios



Deployment lifecycle



Possible upgrades & scalability

Separated instances for PostgreSQL & Redis to get better control over databases:

DB Instances

 PostgreSQL QA

 PostgreSQL Prod

Queues Instances

 Redis QA

 Redis Prod

Additional packages from Laravel ecosystem to implement some functions such as payments:

Laravel Packages

 Laravel Spark

 **Laravel** Dusk

Microservices Architecture

Monolithic architecture of single docker container can be split into microservices. Each container per utility service: auth/payments/indesign etc. Therefore, Kubernetes orchestration system might be implemented for automating deployment, scaling, and management.



Possible Issues

- Microsoft server + InDesign server loading:** Although that jobs will executed in queues, processing of images might overload inDesign server. Server will overloaded exponentially:
$$P = n ^ K$$

P = number of executed synchronous processes, n = number of pharmacies, K = number of entities scheduled for creation. Asynchronous threads (amount of threads depend on server configuration) must be implemented or using Adobe *LBQ (load balancing and queuing server)*.

